

Clarity Scheduling System

Implementation playbook, data-gathering guide, and product structure for a residency scheduling MVP

Purpose
A structured guide for IT, chiefs, coordinators, and leadership to gather the correct scheduling data and build a flexible system across departments.

Who should use this	How to use it
IT / product team	Use as discovery checklist, data model starter, and implementation map.
Chief residents	Use to explain scheduling rules, exceptions, fairness goals, and monthly workflows.
Program leadership	Use to approve scope and understand operational value.

1. Executive Summary

The scheduling system should not be treated as one calendar. It is a layered clinical operations system: resident profiles, annual master schedule, service coverage, call pools, holidays, monthly schedules, resident portal, attendance, and analytics.

Central design principle

Every schedule should be generated from structured resident profiles and configurable department rules, then remain editable by chiefs.

Step	Workflow	Main decision	Owner
1	Annual setup	Blocks, PGY classes, rotations, service minimums, holidays, institutions	Chiefs + IT
2	Resident intake	Master ranking, vacation requests, holidays, fellowship timing, life events, elective choices	Residents
3	Master schedule	Assign each resident to a rotation per block while meeting service minimums	Chiefs
4	Service coverage	Convert master rotations into service rosters by block, service, role, and location	Chiefs
5	Call pool	Pull elective or call-eligible residents into clinic call, jeopardy, cross-cover, no-call, and LOA pools	Chiefs
6	Holiday coverage	Assign holiday staffing using service needs and approved holiday requests	Chiefs
7	Monthly schedules	Build editable service schedules with shifts, calls, clinics, didactics, leave, and coverage checks	Chiefs
8	Publish and track	Resident views, switches, PTO, attendance, statistics, and audit history	Chiefs + Residents

2. Data Taxonomy

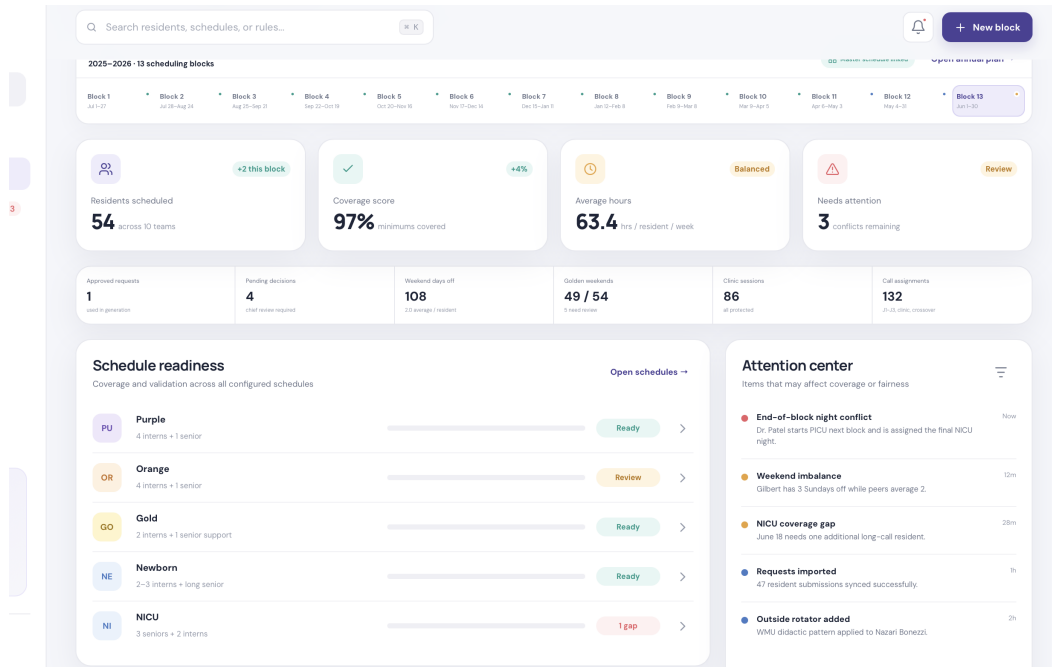
Separate fixed rules, resident-submitted preferences, and generated assignments. This prevents the Excel problem where everything is mixed together and hard to audit.

Data type	Nature	Provided by	Examples
Program structure	Fixed yearly configuration	Chiefs / program leadership	Academic year, number of blocks, block dates, PGY classes, departments, institutions
Resident profile	Mostly fixed, updated as needed	Resident + chiefs	Name, PGY, institution, track, clinic pattern, didactic rules, eligibility flags
Resident requests	Variable, submission-based	Resident	Master preference rank, vacation/PTO, holiday rank, elective rank, fellowship timing, comments
Rotation rules	Fixed per PGY and department	Chiefs	Core rotations, required counts, eligible blocks, vacation-eligible blocks, unsafe transitions
Service rules	Configurable by service	Chiefs with IT support	Service names, roles, staffing min/max, locations, shift types, night rules, call types
Monthly events	Variable by block	Chiefs / coordinator	Holidays, ITE, retreats, conferences, exam days, one-time clinic changes
Generated schedules	System-created, chief-editable	System + chiefs	Master, service rosters, call pools, holidays, monthly service schedules
Operational records	Live transactional data	Residents + chiefs	Call switches, PTO approvals, sick leave, attendance, publish status, notes

3. Product Modules

Module	Purpose	Inputs	Outputs
Master Schedule Builder	Creates the annual source of truth	PGY templates, resident rankings, vacation requests, rotation minimums	Final master grid, service coverage counts, resident master view
Service Coverage Workbook	Shows who covers every service each block	Final master schedule, service mapping rules	Service roster by block, gap/overage alerts, Excel-like view
Call Pool Workbook	Builds eligible call and no-call pools	Elective/call-eligible master blocks, leave, no-call rules	Clinic call, jeopardy, cross-cover, no-call, LOA lists

Module	Purpose	Inputs	Outputs
Holiday Workbook	Plans coverage around holidays	Holiday calendar, resident holiday requests, service minimums	Holiday staffing grid, gap alerts, resident holiday assignments
Monthly Schedule Builder	Builds editable service schedules	Service roster, shift templates, clinics, didactics, PTO, requests	Published service schedules, resident personal schedules, statistics
Resident Portal	Collects requests and shows personal schedule	Published schedules, individual profile, request forms	Requests, call switches, PTO/sick leave, personal statistics
Attendance	Tracks didactic presence	Lecture sessions, resident roster, attendance requirements	Resident attendance history, missed-session counts, chief reports
Rules and Institutions	Stores reusable program logic	Institution profiles, service rules, shift templates, safety rules	Rules automatically applied to future schedules



Chief overview example: readiness, conflicts, and schedule status should be visible without opening multiple files.

4. Department Discovery Process

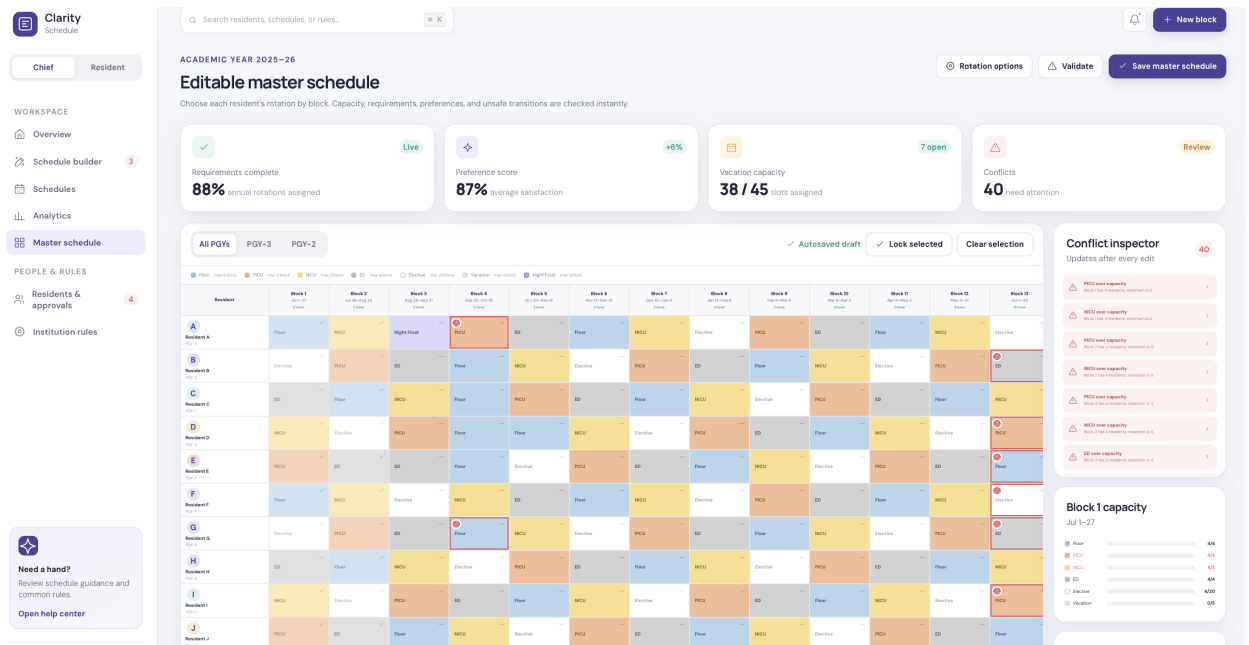
Use this sequence with any new department. It keeps discovery organized and reveals rules that teams often forget to mention.

Interview area	Questions to ask
Program map	How many PGY classes, residents, institutions, tracks, outside rotators, and block systems exist?
Annual master	Which rotations are core, elective, vacation eligible, call eligible, or restricted by PGY year?
Coverage targets	What minimum and maximum number of residents must cover each service per block? Are ranges different by PGY?
Service logic	For each service, what roles exist: day, night, short call, long call, procedure call, unit 1, unit 2, senior, intern?
Protected time	Which clinics, didactics, exams, retreats, conferences, and institution-specific rules must block scheduling?
Requests	What can residents request, what is the deadline, what is first-come-first-serve, and what requires chief discretion?
Fairness	What should be balanced: nights, weekends, golden weekends, holidays, call types, clinic calls, hours, or difficult rotations?
Safety	Which sequences are unsafe: post-night to inpatient, back-to-back calls, call before clinic, max consecutive nights, duty hours?
Outputs	What published PDFs/Excls are expected today, and what digital/Excel-like views must replace them?
Audit	Who can approve, edit, override, publish, view resident-private data, and see other residents' schedules?

5. Master Schedule Builder

The annual master schedule is the source of truth. It assigns each resident to one rotation per block and feeds every downstream schedule.

Configuration area	Chiefs provide	IT builds as options
PGY setup	PGY classes, resident count, number of master schedules needed	PGY selector, count matching, missing-profile alerts
Rotation catalog	Core, elective, vacation-eligible, call-eligible rotations	Editable cards with add, rename, drag, delete, color, service mapping
Coverage targets	Min/max residents per service per block	Coverage counters, shortage/overage warnings
Resident preference intake	Master ranking, vacation, fellowship timing, elective ranking	Annual form, timestamps, sortable ranking queue
Finalize grid	Overrides and final chief decisions	Editable grid, lock/finalize, audit trail



Master schedule example: final grid remains editable while showing coverage warnings.

6. Service Coverage Workbook

Requirement	System behavior
Pull names from master	If a resident is assigned to PICU in Block 2, they appear in the PICU Block 2 service roster.
Support service aliases	A master rotation such as Floor can feed Purple, Orange, Gold, or floor senior depending on rules.
Show counts first	Keep the main view clean with counts and status; reveal names when clicked.
Allow Excel-like view	Chiefs can switch to a familiar spreadsheet-style roster.
Link to profiles	Clicking a resident opens profile and returns chief to the same context.

The screenshot displays the Clarity Scheduling System interface. At the top, there's a navigation bar with tabs for 'Block 1' through 'Block 13'. Below this, the 'Annual scheduling structure' section allows users to set the academic year (2025-2026), number of blocks (13), block model (Four-week blocks), and request deadline (90 days before block start). A sidebar on the left lists steps: Block setup, People, Protected time, Requests, Rules (selected), and Review & generate. The main area shows 'Coverage and fairness rules' for Block 2 (Jul 28-Aug 24). It includes settings for floor team coverage (Minimum interns: 4, Minimum seniors: 1, Max consecutive nights: 5) and weekend fairness (Require one golden weekend: Saturday and Sunday off together). An 'Input readiness' section on the right shows progress for Master schedule linked (54 residents), Resident requests (47 of 47 submitted), Institutional patterns (4 institutions), Outside rotators (6 profiles complete), and Manual events (1 missing end time).

Service builder example: each service needs its own roles, shifts, minimums, and locations.

7. Call Pools, Holidays, and Breaks

Workbook	Who gets pulled in	Chief review questions
Call pool	Elective, call-eligible, or vacation-eligible residents	Who can do clinic call, jeopardy, cross-cover, no-call, or LOA?
Holiday coverage	Available residents filtered by service needs and requests	Are all services covered for each holiday?
Holiday breaks	Residents assigned to Christmas/New Year break windows	Are minimum services covered for both break weeks?

Keep names available without crowding

Use card views and counts by default. When chiefs click a cell or card, show names, source rotation, eligibility reason, and profile links.

8. Monthly Service Schedule Builder

Builder step	What should happen
Block setup	Select block, service, dates, holidays, locations, and source data.
People	Pull residents from master; allow outside rotators and call-pool additions.
Protected time	Auto-pull clinic, didactics, ITE, conferences, PTO, leave, and institution rules.
Requests	Show approved, pending, and denied requests as reminders.
Rules	Apply shift templates, min coverage, eligibility, nights, weekends, and fairness targets.
Review/generate	Create an editable draft, flag conflicts, allow drag/drop, then publish.

Clarity
Schedule

Chief

Resident

WORKSPACE

Overview

Schedule builder

Schedules

Analytics

Master schedule

PEOPLE & RULES

Residents & approvals

Institution rules

Need a hand?

Review schedule guidance and common rules.

Open help center

Q Search residents, schedules, or rules...

K

Block 1 - JUL 1-27, 2026

Schedule workspace

Draft autosaved

Save draft

Export

Publish block

2025-2026 - 13 scheduling blocks

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

Block 8

Block 9

Block 10

Block 11

Block 12

Block 13

Purple

Orange

Gold

Newborn

NICU

PICU

Heme/Onc

Jeopardy

Night Senior

Green Team

Filter

Week 1 - Jun 1-7

Today

Add resident

Day

Night

Protected

Off

Undo

Redo

Purple - Block 1 master-linked roster

Review annual assignments

Coverage plan

High calls

5-5

Evenly distributed

Midweek days covered

4-6

Midweek coverage or holidays

Evening coverage

6 stretches

Evening coverage or holidays

Edit template

View resident totals

Resident

Mon Jun 1

Tue Jun 2

Wed Jun 3

Thu Jun 4

Fri Jun 5

Sat Jun 6

Sun Jun 7

Schedule insights

Resident A

17:00-07:00

17:00-07:00

17:00-07:00

17:00-07:00

17:00-07:00

POST CALL

OFF

Balanced four-week draft

94

High and weekend fairness tracked across the full year

Resident K

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

18:15-07:00

18:15-07:00

Coverage

Need coverage review

Call your residents who are added to unassigned shifts

Weekend coverage

All day and night-time staff

Resident D

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-16:00

06:30-19:00

Fairness

Resident A

94%

High calls

90%

Golden weekends

80%

Resident E

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-19:00

OFF

Resident F

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

06:30-17:00

OFF

Daily coverage

5 / 5

5 / 5

3 / 5

4 / 5

5 / 5

3 / 5

2 / 5

Monthly schedule editor example: generated templates stay editable and show live checks.

9. Resident Portal and Approvals

Resident feature	Purpose	Linked chief-side data
My schedule	Personal work hours, calls, clinics, didactics, PTO, weekends	Published schedules and resident profile
My master schedule	Private annual schedule	Final master schedule
Requests	Days off, weekends, holidays, electives, fellowship notes	Chief approval queue
Call switches	Offer or accept call switch with eligibility checks	Call schedule, clinic rules, back-to-back rules
PTO/sick leave	Submit and track leave	Leave tracker and schedule builder
Attendance	See sessions attended/missed	Attendance module

Q Search residents, schedules, or rules...

K

MY PUBLISHED ASSIGNMENTS

View shared service schedules

Block 13 personal schedule

Only your own assignments are shown here. Open a published service schedule to view the shared team roster.

Expected hours

246 for this block

Calls

4 Clinic, J1, J2, crossover

Weekend days off

4 includes one golden weekend

Current service

Newborn published roster

June 2026

Shift, call, protected time, and rest days

Day Call Protected Off

Jun 1

0630-1700

Day

Jun 2

CLINIC PM

Protected

Jun 3

DIDACTIC

Protected

Jun 4

CLINIC CALL

Call

Jun 5

0630-1700

Day

Jun 6

OFF

Off

Jun 7

OFF

Off

Jun 8

0630-1700

Day

Jun 9

J1 CALL

Call

Jun 10

DIDACTIC

Protected

Jun 11

0630-1700

Day

Jun 12

ITE / OFF

Protected

Jun 13

0630-1800

Day

Jun 14

OFF

Off

Jun 15

0630-1700

Day

Jun 16

CLINIC PM

Protected

Jun 17

J2 NIGHT

Night

Jun 18

POST CALL

Off

Jun 19

0630-1700

Day

Jun 20

OFF

Off

Jun 21

OFF

Off

Clarity Scheduling System Implementation Playbook

Page 6

Resident view example: private personal schedule plus shared published rosters.

10. Requests, Attendance, and Statistics

Area	Required capability
Requests	Approval queue with resident context, system checks, and final chief decision.
Call switches	Check clinic same day/next day, back-to-back calls, and duplicate assignments.
Attendance	Chief creates session; residents check in; profiles update automatically.
Statistics	Coverage, fairness, hours, weekends, golden weekends, calls, leave, attendance.

The screenshot displays the Clarity Scheduling System interface. On the left, a sidebar titled 'PEOPLE & RULES' contains a 'Residents & approvals' section with a count of 4 and an 'Institution rules' section. The main content area is divided into several panels. The top panel shows 'Recurring didactics' (Friday 12:30-17:00) with an 'Automatic' status and an 'Edit override' link. Below this is a 'Call switch approvals' section with a status of '1 awaiting decision'. The bottom panel shows 'Requests awaiting chief decision' with a status of '0 pending'. The interface includes various status indicators, buttons, and a flow diagram for the approval process.

Approvals example: chiefs need request context, system checks, and approve/decline controls.

11. Ownership and MVP Phases

Ownership	Examples
Build once by IT	Authentication, database, import engine, audit logs, permission model, scheduling rule engine, export tools, notification framework
Configure with chiefs	PGY classes, block calendar, service catalog, rotation-to-service mapping, shift templates, min/max staffing, holiday rules, call switch rules
Chiefs manage yearly	Resident roster, outside rotators, approved requests, master schedule decisions, service coverage corrections, holiday staffing
Residents submit	Annual preferences, vacation/PTO, holiday requests, elective rankings, call switch offers/acceptances, sick leave, attendance check-ins
System calculates	Coverage gaps, overages, fairness stats, conflict flags, clinic/didactic conflicts, back-to-back call risk, attendance totals

Phase	Scope
Core MVP	Resident profiles, annual intake, master schedule, service coverage, monthly schedule builder, publishable resident view
Operational MVP	Call pool, PTO/sick leave, call switches, approval queue, attendance check-in, export to Excel/PDF
Advanced MVP	Rule scoring, optimization suggestions, import from legacy Excel/PDF, notifications, audit trail, analytics dashboard
Future expansion	Multi-program support, role-based administration, mobile check-in, integrations with Outlook/Google calendar, institutional reporting

12. Data Model Starter and Acceptance Tests

Object	Key fields
Resident	id, name, pgy, institution, track, clinic_patterns, didactic_rules, leave_balance, eligibility_flags
Block	id, name, start_date, end_date, year, holidays, request_deadline
Rotation	id, name, pgy_allowed, core_required, vacation_eligible, call_eligible, service_mapping, color
Service	id, name, category, locations, roles, shift_templates, min/max staffing, fairness_rules

Object	Key fields
MasterAssignment	resident_id, block_id, rotation_id, source, locked, notes, conflicts
Request	resident_id, type, dates/block, priority, submitted_at, status, decision
ScheduleAssignment	resident_id, service_id, block_id, date, role, shift_template, source, status
AttendanceSession	date, block_id, lecture_type, requirement, allowed_mode, attendees, excused_absences

Minimum end-to-end MVP test

Import a master schedule, generate one service roster, build one monthly service schedule, publish to one resident, submit one call switch, record one attendance session, and export one Excel-like/PDF output.

13. Final Recommendation

Build around source-of-truth layers

Start with resident profiles and master schedule data. Then build service coverage, call pools, holidays, monthly schedules, requests, and attendance as connected layers. This keeps the system explainable, flexible across specialties, and familiar enough for chiefs moving from Excel.